Week 1 Insights Memo: Attenomy Telemetry Anomaly Dataset

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1. Summary

- The dataset has 25,000 service calls and a binary target, **is_anomaly**, which is positive for only 3.9% of rows. It was cleaned and exported reproducibly to Parquet.
- The strongest signal is whether an error was recorded: calls with any error code are anomalous 21% to 24% of the time, against 2.3% for calls with no error. This is also the biggest risk, because the field may reveal the label.
- Every numeric feature is weak on its own (best rank correlation +0.06, best class separation AUC 0.58). Numeric features are not collinear (all pairwise rank correlations are 0.02 or below).
- Week 2 should build a leak-free, stratified pipeline, compare models with and without **error_code**, and judge them by precision-recall metrics, not accuracy.

2. Data and cleaning

- **Checked, nothing found:** no duplicate rows, no malformed or repeated transaction IDs, no unparseable timestamps.
- **Fixed:** channel casing (7 raw values became 4); 809 missing client_tier values became UNKNOWN; 91.8% missing error_code values became NO_ERROR (assumption: missing means no error occurred); 60 negative session durations were set to missing.
- **Left for the training split:** 1,032 missing session durations (4.1%) are filled with the median fitted on training rows only, because the metric is right-skewed.
- **Outliers** are flagged with the IQR rule, not removed (latency reaches 4,546 ms against a median of 71 ms), because extreme values may be the anomalies.
- **Efficiency:** Parquet is 0.69 MB against 2.63 MB for the CSV; memory fell from 3.99 MB to 1.96 MB (51%).
- **Leakage control:** the data is split 70/15/15 (stratified, seed 42) before any fitted step. The imputer, scaler and encoders are fit on training rows only, and all EDA used the training split (17,500 rows, 685 anomalies).

3. Findings

Hypothesis	Test and statistic	Effect size	Result and meaning
H1: anomaly rate depends on error_code	Chi-square, chi2 = 1426.2, dof = 5, p < 1e-300	Cramer's V = 0.285	Rejected. Strong link: 2.3% anomalies with no error, 20.9% to 23.8% with any error.
H2: latency differs by class	Mann-Whitney U = 6,146,454, dof N/A, p = 0.0028	AUC = 0.534	Rejected, but tiny: median 73.8 ms vs 70.5 ms.
H3: anomaly rate depends on client_tier	Chi-square, chi2 = 1.75, dof = 4, p = 0.782	Cramer's V = 0.010	Not rejected. No evidence that the plan level matters.
H4: session duration differs by class	Mann-Whitney U = 5,163,200, dof N/A, p = 0.392	AUC = 0.490	Not rejected. Sessions look the same in both classes.
H5: anomalies have more retries (one-sided)	Mann-Whitney U = 6,679,158, dof N/A, p = 2.5e-17	AUC = 0.580	Rejected. Anomalous calls retry more, a modest effect.

All tests use the training split. The decision threshold is 0.01 (Bonferroni: 0.05 divided by 5 tests). Mann-Whitney was used instead of a t-test because the features are strongly right-skewed (skewness: latency 10.9, payload 5.2, retry count 2.5, session duration 1.4); it compares ranks, so outliers do not distort it. Mann-Whitney has no degrees of freedom (non-parametric). A significant p-value is not enough on its own: with 17,500 rows even tiny differences are significant, so effect sizes are reported.

- The five error types are almost equally risky (20.9% to 23.8%), so what matters is that an error happened, not which one.
- Anomaly rates barely vary by client tier (3.5% to 4.1%) or channel (3.5% to 4.2%). The one hot spot is cli_tool with batch_sync at 7.0% against 3.9% overall; the cell is small, so treat it as a lead to check, not a conclusion.
- Latency outliers are spread evenly: the highest share is 2.0% of calls (channel rest_api, region ap-south-1).

4. Risks

Risk	Evidence	Mitigation
Target imbalance	3.9% positives. A model that always says 'normal' scores 96.1% accuracy.	Stratified splits, class weights, PR-AUC and recall at a fixed precision as the main metrics, threshold tuned on validation only.
Label leakage	error_code gives a 10x lift, while the best numeric feature has AUC 0.58, so models will lean on it. IQR outlier flags were computed on all rows.	Confirm with the data owner that error_code is known before the anomaly decision. Train with and without it. Keep the outlier flags out of the model, or recompute them on training rows.
Collinearity	Low among numeric features (all pairs 0.02 or below). One-hot error codes overlap with NO_ERROR.	Drop one level when encoding, use regularisation, check VIF after encoding. Tree models are unaffected.
Few positives	Only 685 anomalies in training, so results will be noisy.	Stratified 5-fold cross-validation on training rows, report spread, touch the test set once at the end.
Cleaning assumptions	NO_ERROR for missing error_code and median imputation of 4.1% of durations.	Confirm the NO_ERROR meaning with the data owner; test the median choice against a missing-indicator column.

5. Week 2 classical modeling plan

- **Baselines:** a dummy classifier and logistic regression (balanced class weights) on the existing train-only ColumnTransformer.
- **Tree models:** random forest and histogram gradient boosting, tuned with stratified cross-validation on training rows.
- **Leakage ablation:** compare all models with error_code, with only a has_error flag, and without it, to measure how much performance depends on it.
- **Feature work (fit on training rows only):** hour and weekday from the timestamp, log transforms of skewed features, outlier flags recomputed on training rows, and per-account history features that use only earlier timestamps.
- **Evaluation:** PR-AUC as the primary metric, plus recall at a precision target agreed with the mentor, F1 and the confusion matrix on validation. Save the winning pipeline in models/ and add tests for any new functions.

Questions for review: (1) Should clean.parquet itself have zero nulls, or is filling them after the split acceptable? (2) Is error_code known before the anomaly label is set? (3) Is NO_ERROR the right reading of a missing error_code?